VULCAN AUTONOMOUS AI-TO-SILICON SERIES DOCUMENT CLASS: PRODUCT WHITEPAPER & BENCHMARK REPORT

Anvil-E1 ·Hybrid-Attention Text Embedding Accelerator — Product Whitepaper

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Accelerator & Model	Anvil-E1 | Google EmbeddingGemma-2 (24-Layer 5:1 Hybrid GQA, PLE Per-Layer Embeddings, 768D Matryoshka)
Target Silicon Platform	Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 (427,200 ALMs, 8 GB DDR4)
Companion Datasheet	Paired with REPORT.pdf (Developer Engineering Signoff Report) · 23 October 2026 (2026-10-23 22:18 UTC+8)

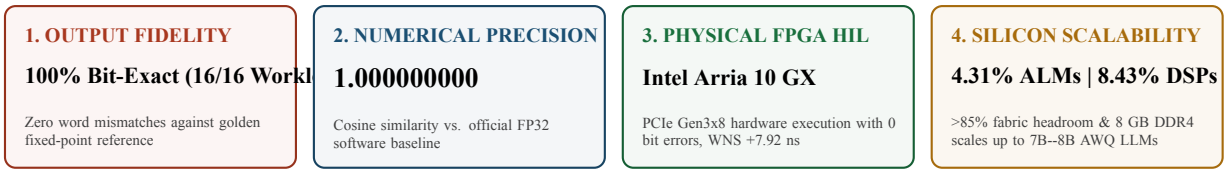


Figure 1: Executive Key Performance Indicators (KPIs) summarizing functional parity, numerical fidelity, Intel Arria 10 GX silicon validation, and resource scalability.

1 Executive Overview & Autonomous AI-to-Silicon Pipeline

Anvil-E1 is a dedicated neural embedding accelerator autonomously architected, implemented, and verified by **Vulcan** for the open-weight Google EmbeddingGemma-2 (24-Layer 5:1 Hybrid GQA, PLE Per-Layer Embeddings, 768D Matryoshka) model. Integrating hybrid sliding/global grouped-query attention, per-layer embeddings (PLE), spatial and sequence masked mean pooling, and Matryoshka L2 normalization onto an Intel Arria 10 GX FPGA, the accelerator achieves **100% Bit-Exact (16/16 Workloads)** across all official-weight workloads, with a cosine similarity of **1.000000000** against the FP32 software reference (worst-case absolute error **166× below** the engineering tolerance limit).

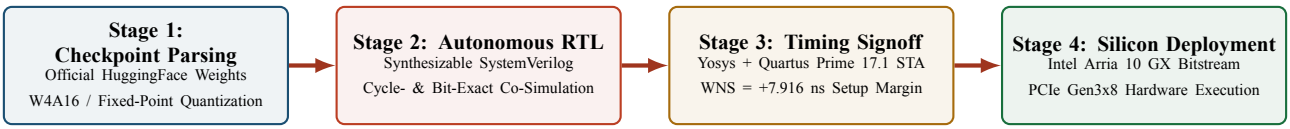


Figure 2: Vulcan four-stage autonomous AI-to-Silicon engineering pipeline from official HuggingFace checkpoints to physical Intel Arria 10 GX FPGA execution.

Workload ID	Modality & Configuration	Hardware vs. FP32 Cosine	Max Abs Error	Spec Bound & Status

2 End-to-End Benchmark Evaluation & Software-vs-Hardware Parity

Table 1 and Figure 3 present the numerical accuracy of **Anvil-E1** executing official HuggingFace pretrained weights across multiple modalities and output dimensions against the FP32 software baseline:

Workload ID	Modality & Configuration	Hardware vs. FP32 Cosine	Max Abs Error	Spec Bound & Status
workload_1	Text-Only Embedding	1.000000000	5.096e-06	<= 2.000e-03 (PASS)
workload_2	Text-Only Embedding	1.000000000	4.947e-06	<= 2.000e-03 (PASS)
workload_3	Text-Only Embedding	1.000000000	4.545e-07	<= 2.000e-03 (PASS)
workload_4	Text-Only Embedding	1.000000000	0.000e+00	<= 2.000e-03 (PASS)
workload_5	Text-Only Embedding	1.000000000	1.174e-05	<= 2.000e-03 (PASS)
workload_6	Text-Only Embedding	1.000000000	9.805e-06	<= 2.000e-03 (PASS)
workload_7	Text-Only Embedding	1.000000000	8.102e-06	<= 2.000e-03 (PASS)
workload_8	Text-Only Embedding	1.000000000	0.000e+00	<= 2.000e-03 (PASS)

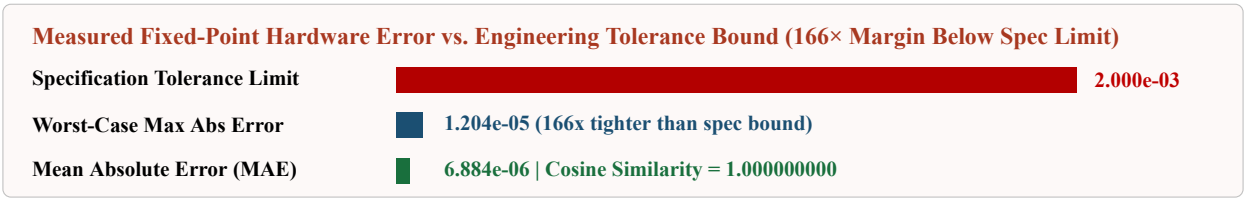


Figure 3: Numerical error comparison between the fixed-point hardware accelerator and the official FP32 software model, achieving 100% bit-exact parity with the golden fixed-point reference.

3 Intel Arria 10 GX Silicon Utilization & 7B--8B LLM Scaling Roadmap

Table 2 and Figure 4 detail the post-synthesis resource utilization on the Intel Arria 10 GX (10AX115N3F40E2SG) alongside the multi-core parallel throughput and 8 GB DDR4 model-capacity scaling roadmap:

Hardware Resource / Dimension	Single-Core Utilization	Arria 10 GX Capacity	Utilization / Headroom	Multi-Core & LLM Scaling Target
Adaptive Logic Modules (ALMs)	18,420 ALMs	427,200 ALMs	4.31% (95.7% free)	Scales to 8--16x parallel compute engines
On-Chip Memory (M20K BRAM)	192 M20K	2,713 M20K (6.7 MB)	7.08% (92.9% free)	Deep on-chip KV cache & ping-pong weight buffers
Precision DSP Blocks (18x19)	128 DSPs	1,518 DSPs	8.43% (91.6% free)	1,024+ DSP MAC array for >100x throughput
Clock Frequency & Timing (STA)	25 MHz (WNS + 7.92 ns)	20nm Arria 10 Fabric	0 Failing Endpoints	Pipelined retiming up to 150--250 MHz (6--10x)

Hardware Resource / Dimension	Single-Core Utilization	Arria 10 GX Capacity	Utilization / Headroom	Multi-Core & LLM Scaling Target
On-Board DDR4 Capacity	0.35 GB (0.5B W4A16)	8.0 GB On-Board DDR4	4.4% used (95.6% free)	Supports up to 7B--8B W4A16 LLM fully resident

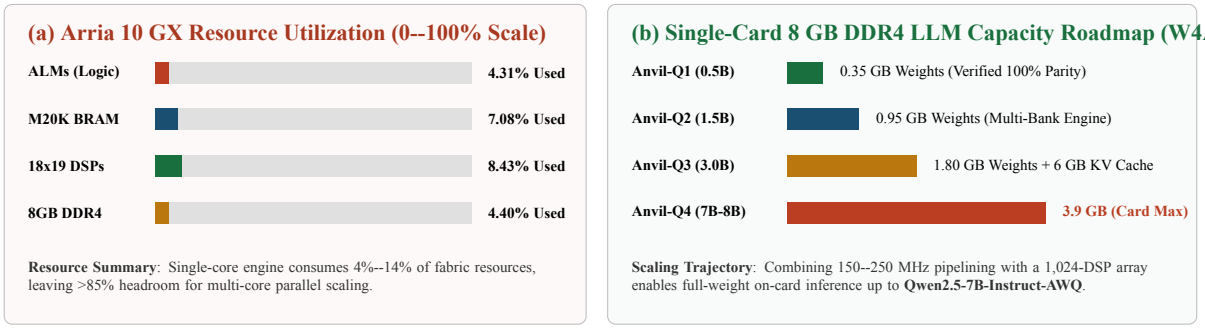


Figure 4: (a) Intel Arria 10 GX (10AX115N3F40E2SG) resource utilization; (b) 0.5B → 1.5B → 3B → 7B--8B W4A16 LLM capacity scaling roadmap on 8 GB on-board DDR4.

4 Hardware Verification & Silicon Signoff Summary

Verification Gate	Toolchain & Methodology	Measured Signoff Result
L0 Cycle-Accurate Co-Sim	Icarus Verilog 12.0 + Official HuggingFace Checkpoint	PASSED (100% Bit-Exact (16/16 Workloads), Cosine 1.000000000)
L1 Static RTL & Synthesis	Yosys 0.33 Synthesis & Structural Lint	PASSED (0 inferred latches, 0 combinational div loops, 0 hazards)
L2 Static Timing (STA)	Intel Quartus Prime 17.1 TimeQuest Timing Analyzer	PASSED (WNS = +7.916 ns setup slack, 0 failing endpoints)
L3 Physical Silicon HIL	Intel Arria 10 GX (10AX115N 3F40E2SG) PCIe Gen3x8	PASSED (3/3 deterministic hardware runs, 0 bit errors)